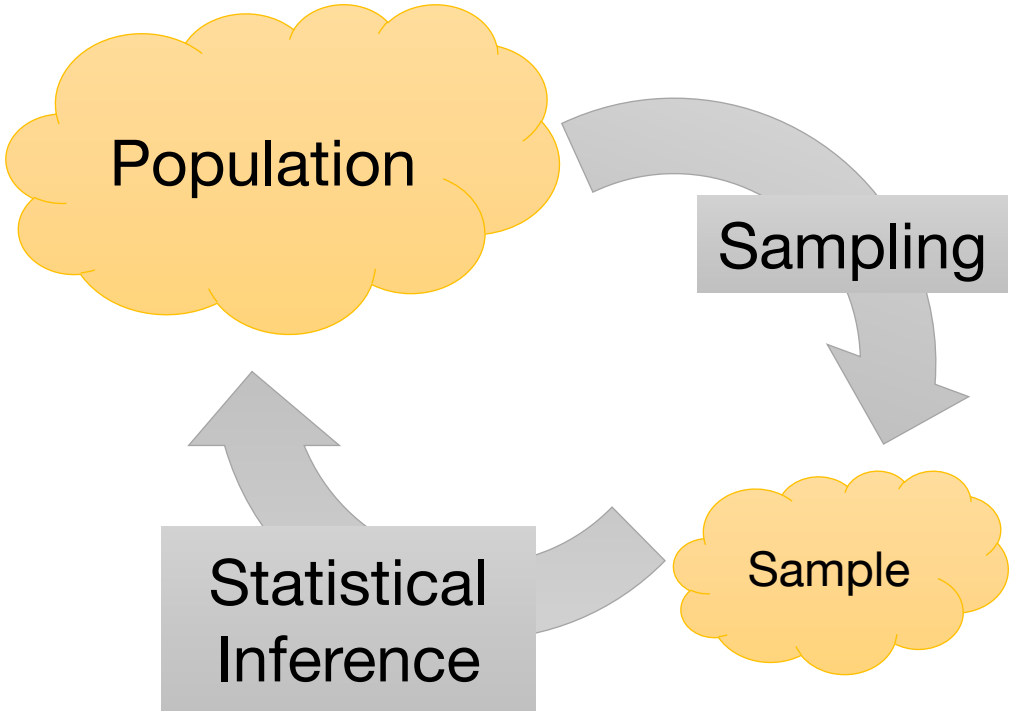
Lecture #11 | Hypothesis Testing

SE377 Introduction to Big Data Analysis and Visualization (2017)

Prof. Min-gyu Cho

Population, a sample and statistical inference

- **Population**: all individual or objects of interest
- **Sample**: all the cases that we have collected data on (a subset of the population)
- **Statistical inference**: the process of using data from a sample or samples to gain information about the population



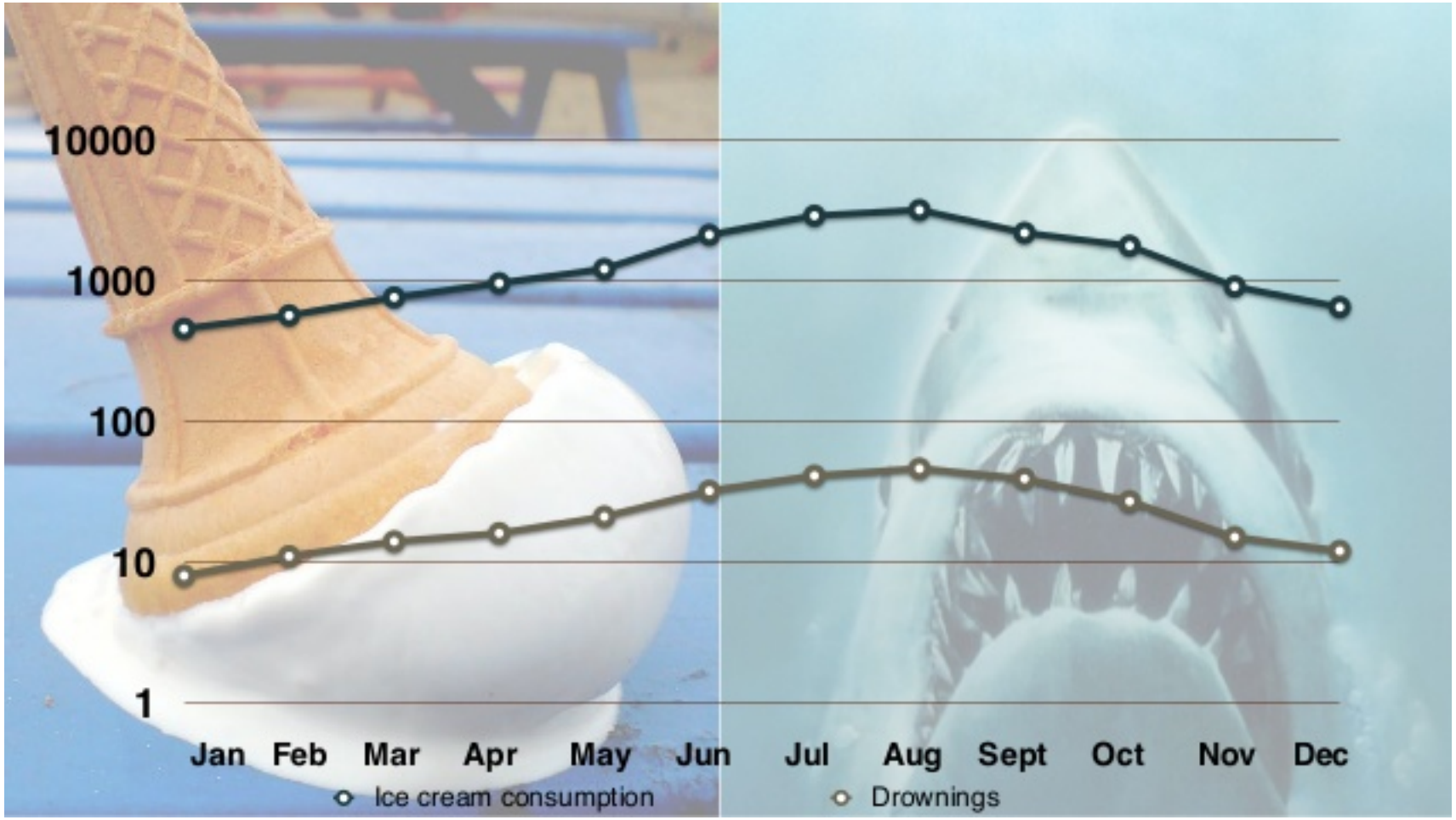
Sampling bias

- Sampling bias occurs when the method of selecting a sample causes the sample to differ from the population in some relevant way
- If sampling bias exists, we cannot trust generalizations from the sample to the population
- ***"Random sampling"*** is important to avoid sampling bias
  - In many cases, taking "random sample" can be very difficult
  - Usually taking the same ratio of data is better than taking the same number of observations from a certain data types

Bias

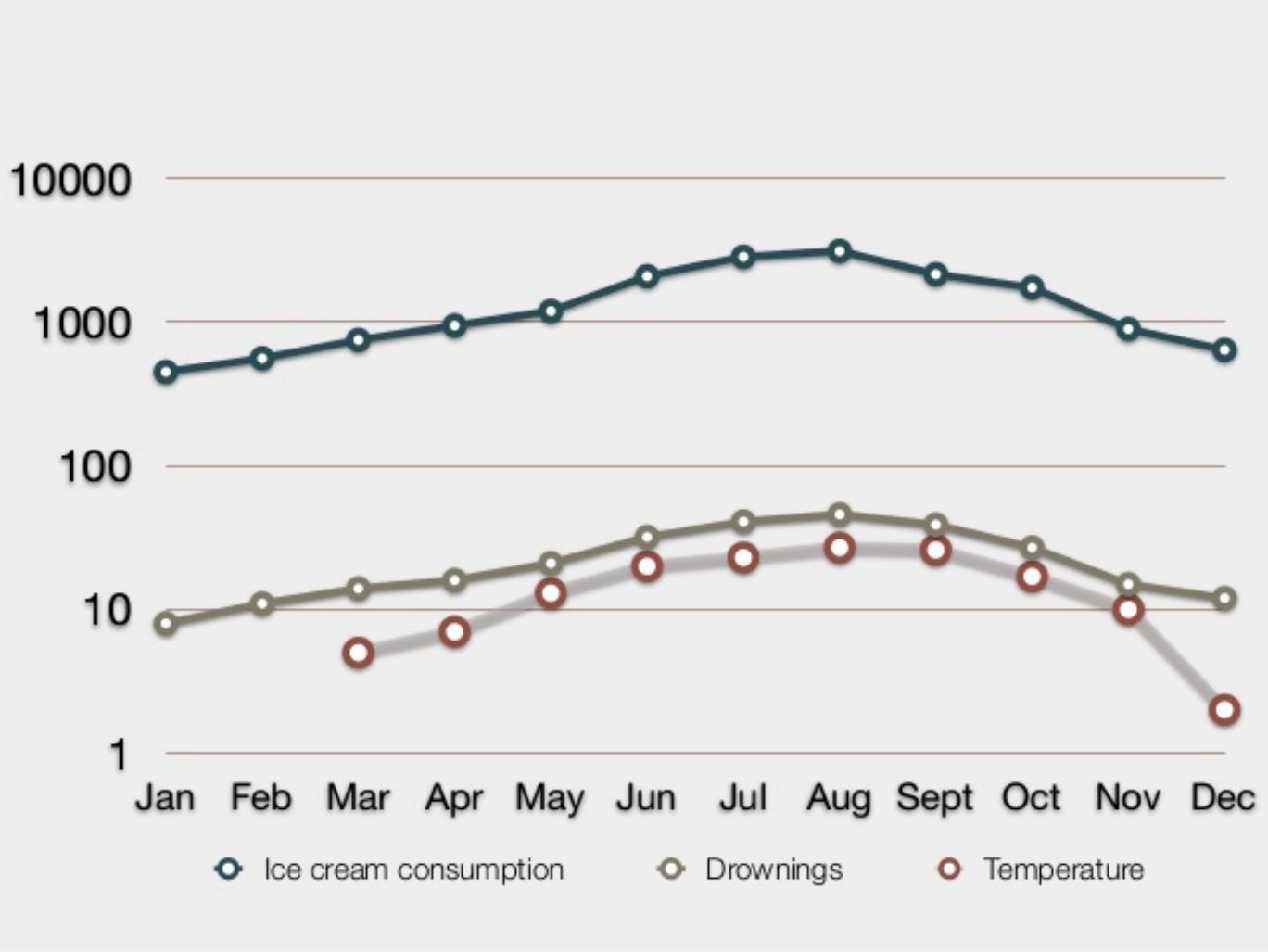
- You should be ***"critical"*** when handling data to eliminate any forms of bias!!!
- Types of bias (not exaustive)
  - Sampling bias
  - Question wording
  - Context or Conditional probability
  - Inaccurate responses
    - Self-confidence (e.g., 95% US drivers think their driving skills are above the average)
    - Placebo effects
    - …

One more example on correlation v.s. causality



Source: Lean Analytics

Confounding variable



Example: kidney stone

- There are two types of treatments in handling kidney stone
- Which treatment is better?

	Success	Failure	Success rate
Treatment A	273	77	78%
Treatment B	289	61	83%

Example: kidney stone (cont.)

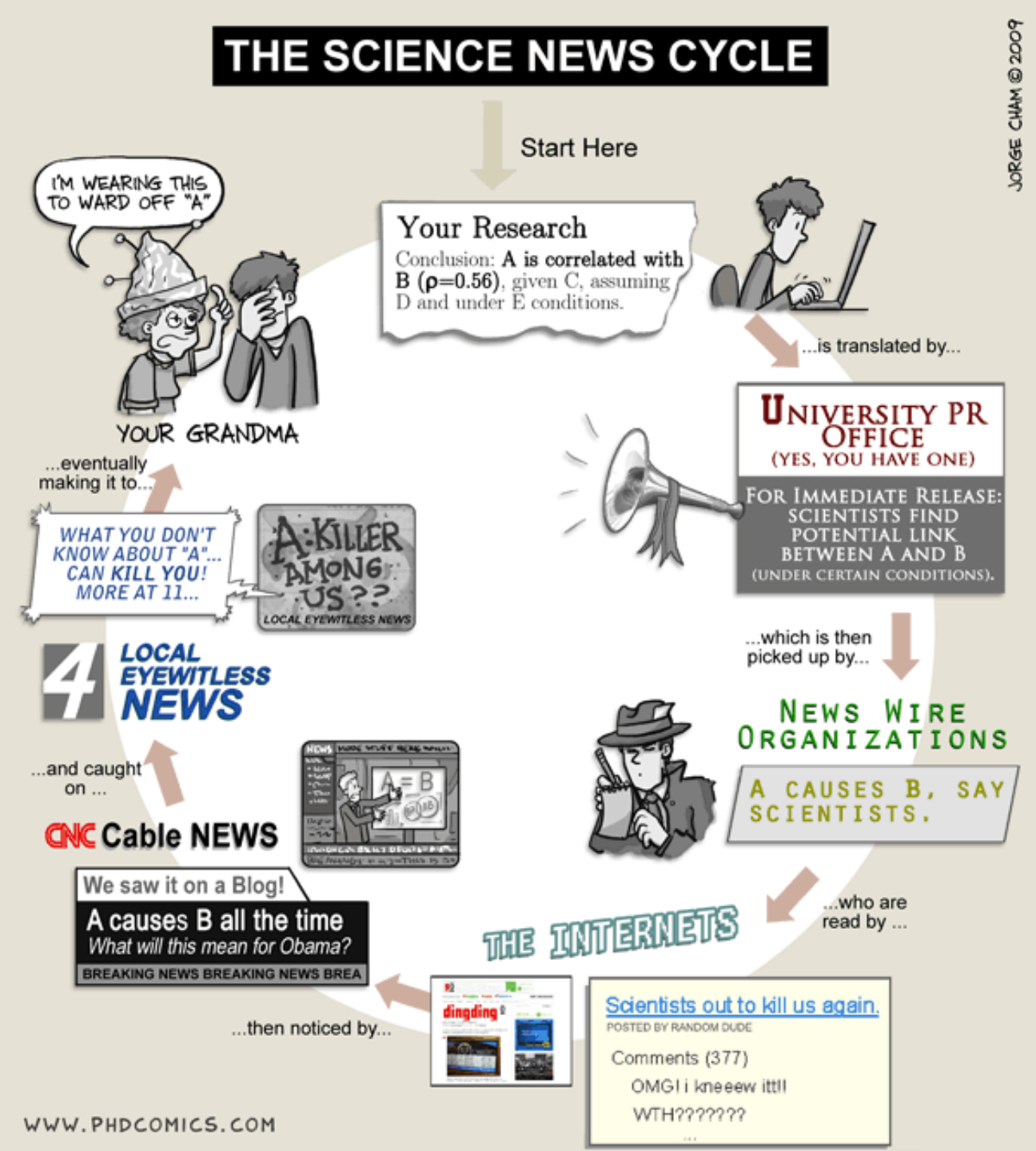
All stones	Success	Failure	Success rate
Treatment A	273	77	78%
Treatment B	289	61	83%

Small stones	Success	Failure	Success rate
Treatment A	81	6	93%
Treatment B	234	36	87%

Large stones	Success	Failure	Success rate
Treatment A	192	71	73%
Treatment B	55	25	69%

Experiment v.s. Observational study

- An ***"experiment"*** is a study in which the researcher actively controls one or more of the explanatory variables
- An ***"observational study"*** is a study in which the researcher does not actively control the value of any variable, but simply observes the values as they naturally exist
  - Observational studies can almost never be used to establish causation


THE SCIENCE NEWS CYCLE
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Start Here
Your Research
Conclusion: A is correlated with B (ρ=0.56), given C, assuming D and under E conditions.
...is translated by...
UNIVERSITY PR OFFICE
(YES, YOU HAVE ONE)
FOR IMMEDIATE RELEASE: SCIENTISTS FIND POTENTIAL LINK BETWEEN A AND B
(UNDER CERTAIN CONDITIONS).
...which is then picked up by...
NEWS WIRE ORGANIZATIONS
A CAUSES B, SAY SCIENTISTS.
...who are read by ...
THE INTERNETS
Scientists out to kill us again.
POSTED BY RANDOM DUDE
Comments (377)
OMG! i kneeew itt!!
WTH???????
...then noticed by...
We saw it on a Blog!
A causes B all the time
What will this mean for Obama?
BREAKING NEWS BREAKING NEWS BREA
...and caught on ...
CNC Cable NEWS
4 LOCAL EYEWITLESS NEWS
A=B
A KILLER AMONG US??
LOCAL EYEWITLESS NEWS
WHAT YOU DON'T KNOW ABOUT "A"... CAN KILL YOU! MORE AT 11...
...eventually making it to...
YOUR GRANDMA
I'M WEARING THIS TO WARD OFF "A"
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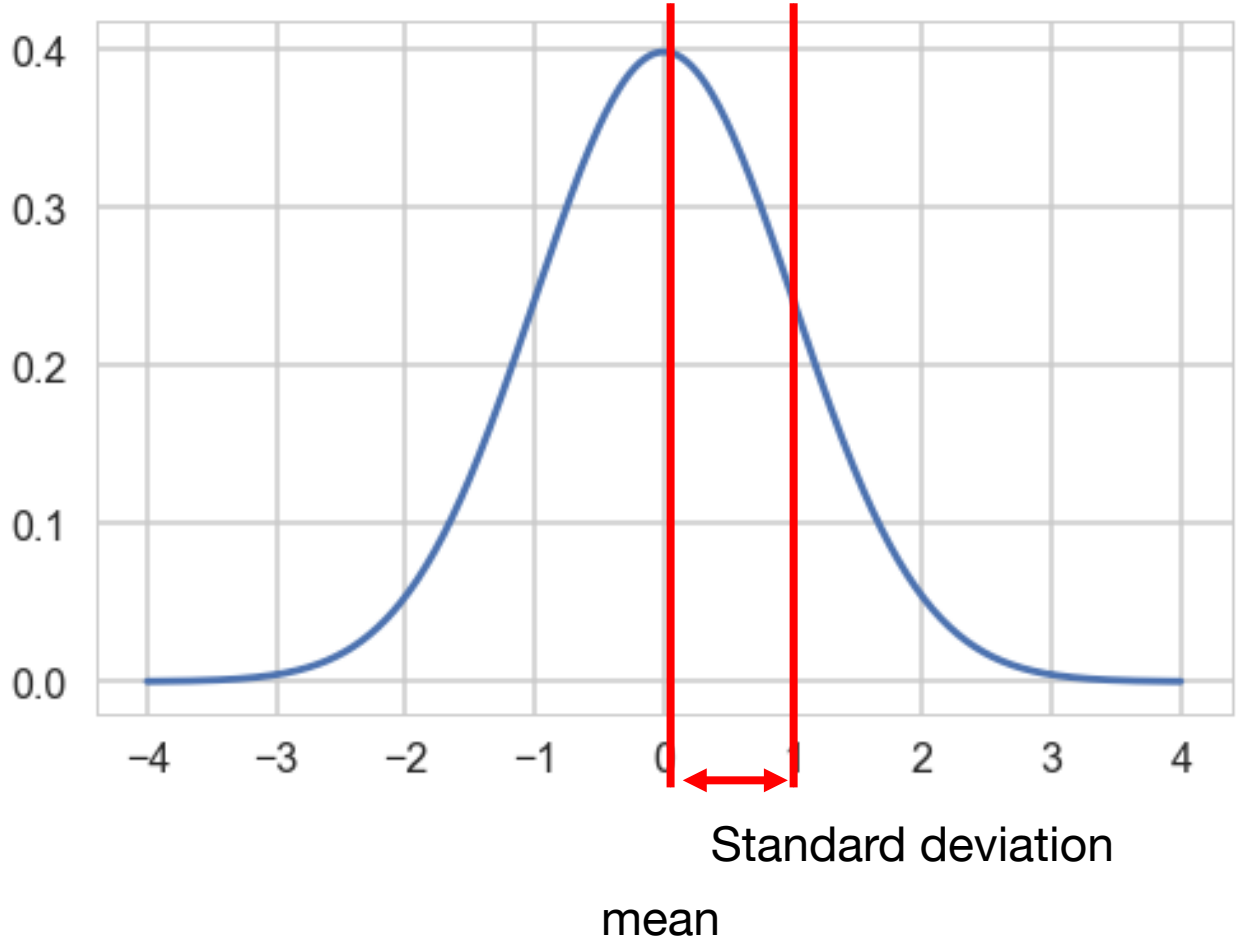
Distributions

- Normal: the most popular and basic distribution
- Poisson: the distribution of counts that occur at a certain “rate”.
  - Observed frequency of a given term in a corpus.
  - Number of visits to a web site in a fixed time interval.
  - Number of web site clicks in an hour.
- Exponential: the interval between two such events.
- Zipf/Pareto/Yule distributions: govern the frequencies of different terms in a document, or web site visits.
- Binomial/Multinomial: The number of counts of events (e.g. die tosses = 6) out of n trials.

Normal distribution

- pdf: $\mathrm{f}(\mathrm{x}|\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)\text{^}2}{2\sigma^2}}$
- Mean: center
- Variance: a measure of width of distribution
- Standard deviation: square root of variance
- Completely characterized by mean and variance
- Mean ± 2 std. dev. ~ 95%

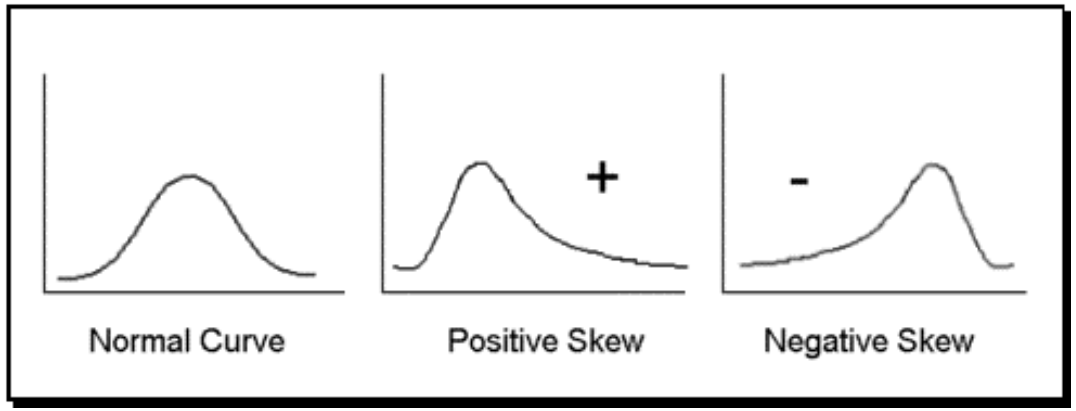


Central Limit Theorem

- The distribution of the sum (or mean) of a set of n identically-distributed random variables X_i approaches a normal distribution as $n \to \infty$.
- The common parametric statistical tests, like t-test and ANOVA assume normally-distributed data, but depend on sample mean and variance measures of the data.
- They typically work reasonably well for data that are not normally distributed as long as the samples are not too small.

Correcting distribution

- Many statistical tools (e.g., t-test, ANOVA, mean/variance) assume normal distribution, but in many cases, this assumption is NOT valid
- You can check the distribution with box plot or histogram
- Examples
  - If X satisfies a log-normal distribution, Y= log(X) has a normal distribution
  - If X is a Poisson distribution with mean k and standard deviation sqrt(k), sqrt(X) is approximately normally distributed with standard deviation 1

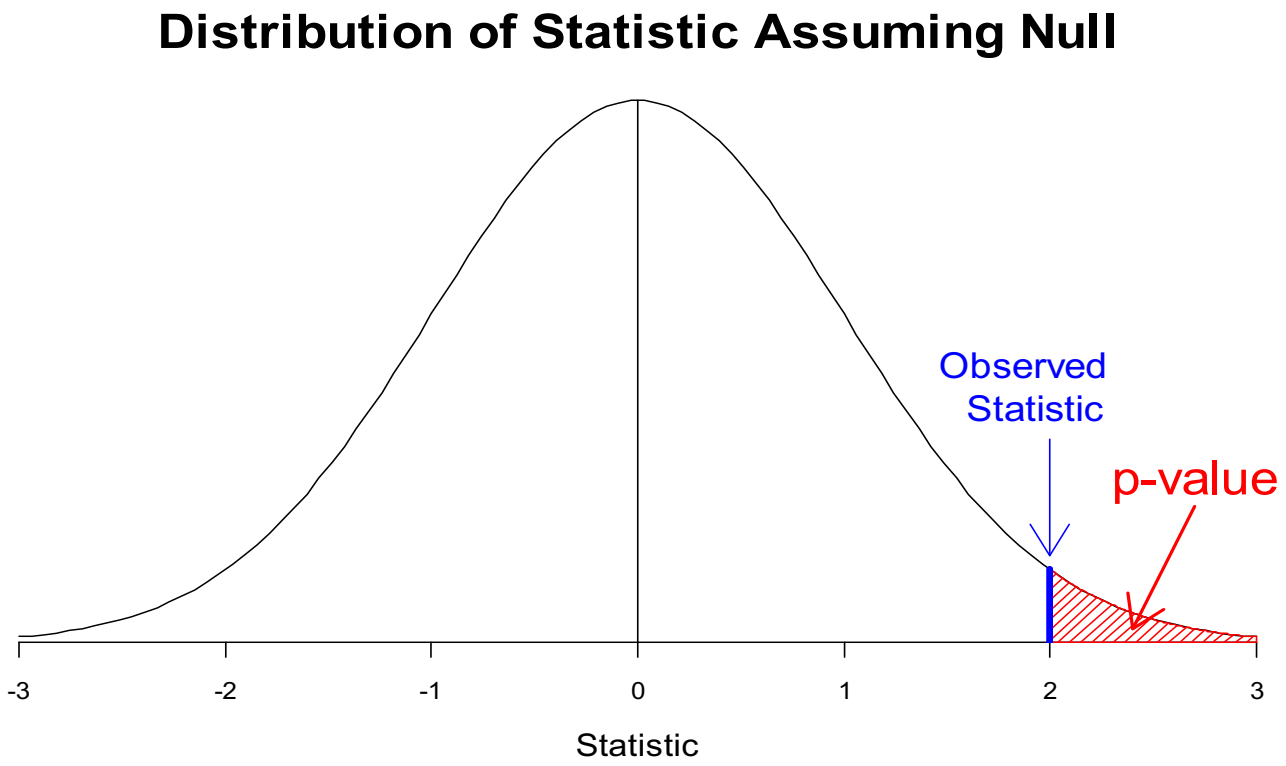


Statistical Hypotheses

- Statistical tests are framed formally in terms of two competing hypotheses:
  - **Null Hypothesis** (H_0): Claim that there is no effect or difference.
  - **Alternative Hypothesis** (H_a): Claim for which we seek evidence.

Steps in hypothesis testing

1. State Hypotheses
2. Calculate a test statistic, based on your sample data
3. Create a distribution of this test statistic, as it would be observed if the null hypothesis were true
4. Use this distribution to measure how extreme your test statistic is



Formal decisions

- If the p-value is small:
  - REJECT H_0
  - The sample would be extreme if H_0 were true
  - The results are statistically significant
  - **<u>We have evidence for H_a</u>**
- If the p-value is not small:
  - DO NOT REJECT H_0 (This does NOT mean accept H_0)
  - The sample would not be too extreme if H_0 were true
  - The results are not statistically significant
  - **<u>The test is inconclusive; either H_0 or H_a may be true</u>**

Errors

		Decision	
		Retain the null	**Reject the null**
Truth in the population	True	CORRECT $1 - \alpha$	TYPE I ERROR α
	False	TYPE II ERROR β	CORRECT $1 - \beta$ POWER

One sample T-test

1. The **one-sample t-test** is used to determine whether a **sample** comes from a population with a specific mean. This population mean is not always known, but is sometimes hypothesized.

2. Compute the test statistics

$$t = \frac{\bar{X} - A}{\bar{\sigma}}$$

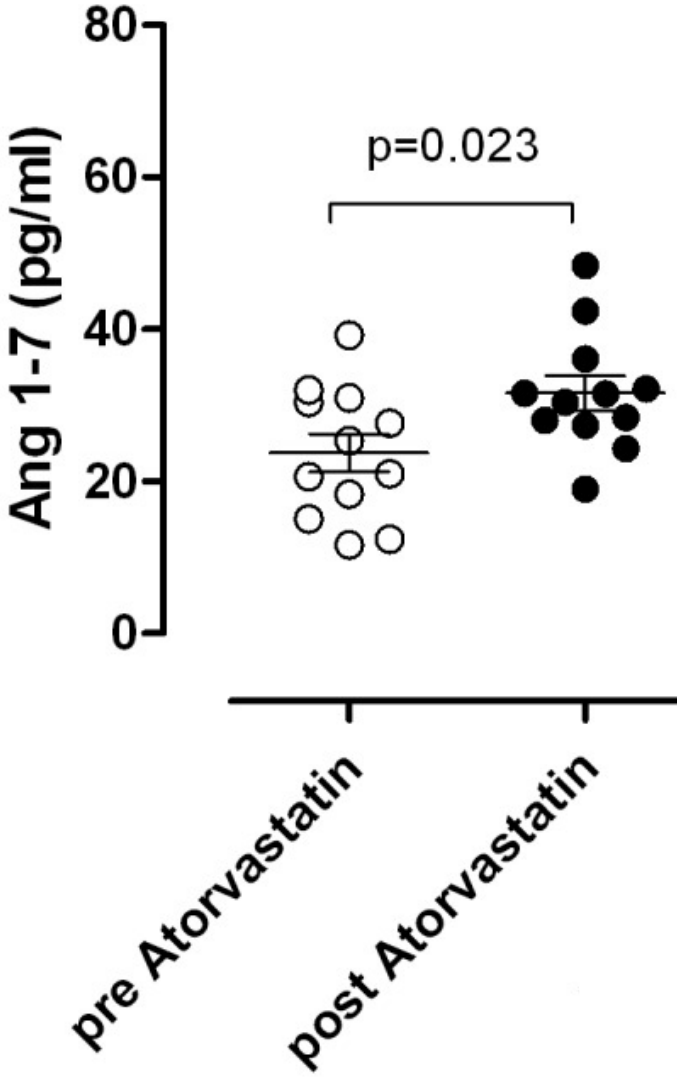
where $\bar{X}$ is the sample mean, $\bar{\sigma}$ is the sample standard deviation, which is the square root of the sample variance Var(X), and A is the hypothesized mean.

Two sample T-test

1. In this test, there are two samples X_1 and X_2. A t statistic is constructed from their sample means and sample standard deviations:

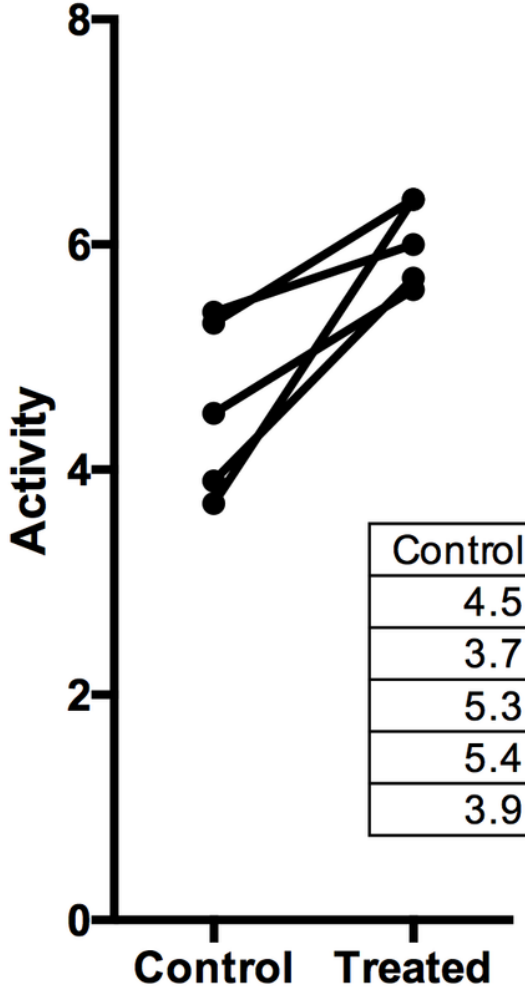
$$t = \frac{\overline{X_1} - \overline{X_2}}{s_{\overline{X_1}-\overline{X_2}}}, \text{where } s_{\overline{X_1}-\overline{X_2}} = \sqrt{\frac{s_1^2}{n_1} - \frac{s_2^2}{n_2}}$$

2. You should try to understand the formula, but you shouldn't need to use it. (Most stat software exposes a function that takes the samples X_1 and X_2 as inputs directly.)



Paired T-test

- You use the single-sample test for one group of individuals in two conditions. Just subtract the two measurements for each subject, and use the difference for the t-test.



Control	Treated
4.5	5.6
3.7	6.4
5.3	6.4
5.4	6.0
3.9	5.7

Chi-square test for goodness of fit

1. State null hypothesized proportions for each category, p_i. Alternative is that at least one of the proportions is different than specified in the null.

2. Calculate the expected counts for each cell as np_i . Make sure they are all greater than 5 to proceed.

3. Calculate the χ^2 statistic: $$\chi^2 = \sum \frac{(\text{observed - expected})^2}{\text{expected}}$$

4. Compute the p-value as the area in the tail above the χ^2 statistic, for a χ^2 distribution with df = (# of categories – 1)

5. Interpret the p-value in context.

Chi-square test for association

1. H_0: The two variables are not associated
   H_a: The two variables are associated
2. Calculate the expected counts for each cell
   $$\text{expected} = \frac{\text{row total} \times \text{column total}}{\text{sample size}}$$
   Make sure they are all greater than 5 to proceed.
3. Calculate the χ^2 statistic
   $$\chi^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}}$$
4. Compute the p-value as the area in the tail above the χ^2 statistic, for a χ^2 distribution with $df = (r-1)\times(c-1)$
5. Interpret the p-value in context.

Analysis of Variance (ANOVA)

- Analysis of Variance (ANOVA) compares the variability between groups to the variability within groups

Total Variability = **Variability Between Groups** + **Variability Within Groups**

ANOVA Table

Source	df	Sum of Squares	Mean Square	F Statistic	p-value
Groups	k-1	SSG	MSG = SSG/(k-1)	$\frac{\text{MSG}}{\text{MSE}}$	Use $F_{k-1,n-k}$
Error	n-k	SSE	MSE = SSE/(n-k)		
Total	n-1	SST			

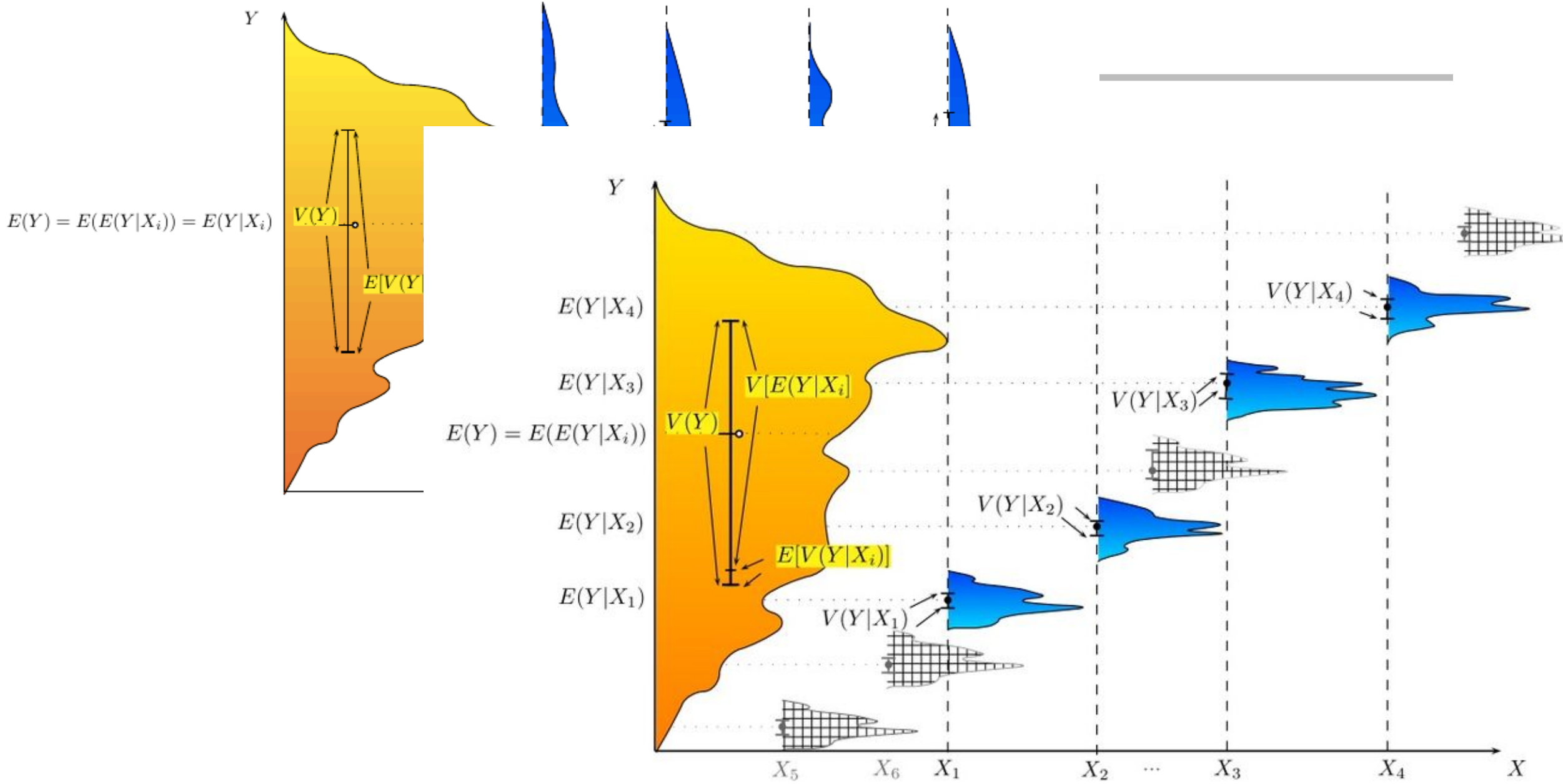


Figure 3: ANOVA : very good fit

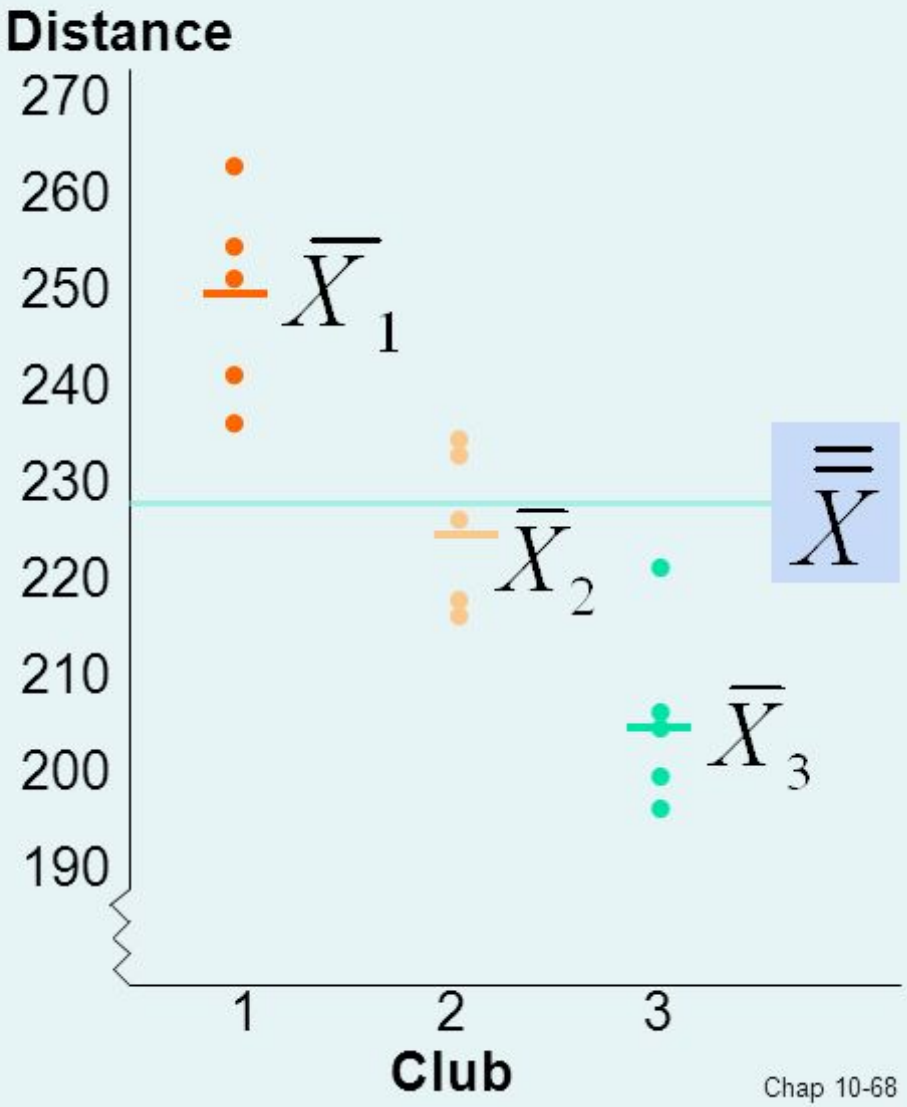
One-Way ANOVA Example: Scatter Plot

DCOVA

Club 1	Club 2	Club 3
254	234	200
263	218	222
241	235	197
237	227	206
251	216	204

$\bar{X}_1 = 249.2$	$\bar{X}_2 = 226.0$	$\bar{X}_3 = 205.8$

$\bar{\bar{X}} = 227.0$



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Reading List

- Inference methods: http://prezi.com/c1xz1on-p4eb/stat-101/

ANY QUESTIONS?